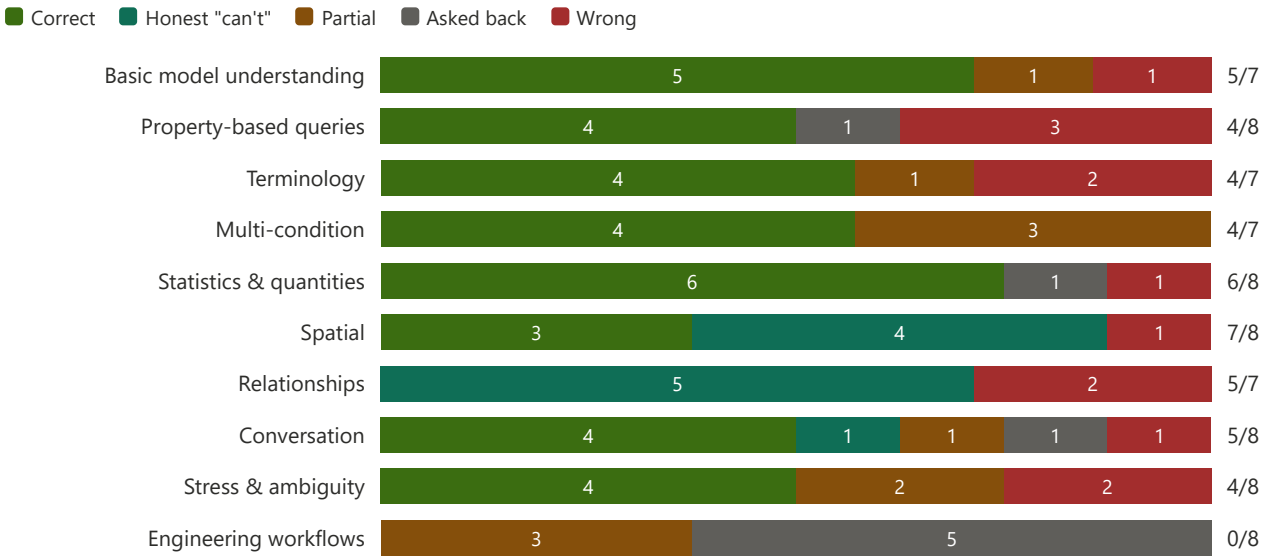
Augmented: test report

76-question test plan on **rstbasicsampleproject.rvt** (845 elements) · real OpenAI (gpt-6-luna) · run 2026-11-07 01:02 UTC · same code as the live app (V6.2.0, local run)

58% pass: correct or honest "can't" (44/76)	**45%** correct answers (34/76)	**13** wrong answers (17%)	**0** invented elements or numbers
198,186 tokens in total (≈ $0.025)	**2,608** tokens per question on average	**2.8 s** median answer time (90%: 4.7 s)	**8** asked back instead of answering

In one sentence: Augmented never invented an element or a number: 34 answers were fully correct and 10 honestly said the data needed isn't available, while the 13 wrong answers come from a few repeatable causes, led by one word-matching bug ("structural") that alone accounts for 5 of them.

Results by section



Section	Correct	Honest "can't"	Partial	Asked back	Wrong	Pass	Tokens/q
Basic model understanding	5		1		1	**71%**	698
Property-based queries	4			1	3	**50%**	2,677
Terminology	4		1		2	**57%**	1,149
Multi-condition	4		3			**57%**	4,223
Statistics & quantities	6			1	1	**75%**	1,698
Spatial	3	4			1	**88%**	2,943
Relationships		5			2	**71%**	3,104
Conversation	4	1	1	1	1	**63%**	2,465
Stress & ambiguity	4		2		2	**50%**	3,571
Engineering workflows			3	5		**0%**	3,393

How the grading works

Correct: the answer matches the model data (every number was checked with SQL against the model's own database). **Honest "can't"**: the question needs data the app doesn't have (grids, connectivity, viewer selection, clash geometry) and the app said so without inventing anything; this counts as a pass. **Partial**: right direction, but incomplete or with a misleading remark. **Asked back**: the app asked a clarifying question where it could have answered. **Wrong**: a wrong number, a misleading list, or a refusal of a valid question.

Pass rate = (correct + honest) / all questions. "Invented" means elements or values that don't exist in the model; none were found. Two answers (R02, R06) were misleading: they presented a level filter as a structural relationship.

Cost and speed

Answered by code only (tier 0, incl. the ~150-token intent check)	20 questions, 131 tokens on average	13 correct
Tier 1 (candidates prompt)	25 questions, 1,221 tokens on average	12 correct
Tier 2 (full model vocabulary)	31 questions, 5,324 tokens on average	9 correct
Whole run	198,186 tokens ≈ $0.025 at gpt-6-luna prices	median 2.8 s, slowest 8.0 s

This plan is deliberately hard: about a third of it asks for capabilities outside the app's data (relationships, grids, selection). Typical tester questions are cheaper; the earlier everyday-question sets were 89% answered by code at 0-150 tokens.

What to fix, in order of impact

#	Cause	Questions	Impact	Fix
1	**The word "structural" is matched to the family "Structural Area Reinforcement"**	P08, N03, A08, S08, E08 (+ A07 cost 5k tokens)	5 wrong answers	Code (matcher): free to fix and test
2	**Numeric filters on a property the category doesn't have, or in another unit**	P01 (walls: Width, not Thickness), E04 (widths in cm), earlier Q10	2 wrong answers that look like real "0"s	Code (engine): say "no such property", convert cm/mm
3	**Relationship questions answered with a level filter**	R02, R06	2 misleading answers	Prompt/validator: "supports/connected" -> "not in the data"
4	**Missing values not used ("missing Mark", "no material")**	P05, W02, W03, C04	4 asked back	Planner: use the existing "no value" filter
5	**Single words make tier 2 refuse the question**	P07 ("whose"), N05 ("isolated")	2 wrong refusals	Planner: treat such words as filler
6	**Follow-up context lost on "How many elements remain?"**	I04	1 wrong	Follow-up detection: "remain" = count the last result
7	**Levels are not a known question type**	B05	1 wrong	Add "list the levels" (values of the level properties)
8	**Placeholder "<By Category>" shown as a material**	B06	1 partial	Hide placeholders in group results
9	**Multi-column reports (inventory, schedule with length/material)**	W04, W06, W01	3 partial	Feature: chosen columns in the result table + CSV export

Fixes 1, 2, 7 and 8 are deterministic code changes that can be verified for free with the dry run. Estimated effect of fixes 1-6: about 15 of the 21 wrong/asked-back answers become correct, taking the pass rate from 58% to roughly 78%.

Every question, answer and verdict

Basic model understanding

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
B01	How many elements are in this Revit model?	There are 845 elements. Showing the first 500.	845 elements (all real Revit elements, family/type nodes excluded).	Correct	1,050 tier 1
B02	What are the different categories of elements in this model?	elements by category: • Structural Framing: 370 • Structural Rebar: 346 • Structural Area Reinforcement: 44 • Structural Foundations: 36 • Structural Columns: 30 • Walls: 9 • Floors: 5 • Supports: 2 • Lines: 1 • Runs: 1 • Stairs: …	11 categories; Structural Framing 370, Rebar 346, Area Reinforcement 44, Foundations 36, Columns 30, Walls 9, Floors 5 …	Correct	1,058 tier 1
B03	How many structural columns are in the model?	There are 30 Structural Columns elements.	30 structural columns.	Correct	154 tier 0
B04	List all the structural framing types used in this model.	Structural Framing elements by Type: • Timber_50 x200: 106 • SC15012: 96 • Timber_50 x150 _ContentGenerator: 65 • 200UB25.4: 35 • 356mm: 27 • 88.9x5.0CHS: 18 • 30mm: 16 • 200PFC: 7	Framing types with counts (Timber_50 x200 106, SC15012 96, …).	Correct	156 tier 0
B05	What are the different levels defined in this project?	Found 845 elements. Showing the first 500.	7 levels: Foundation, Level 1, Level 2, Level Lower, Ref Beam system, Top of Parapet, Ceiling (elevations not in the data). Listed all 845 elements instead of the levels.	Wrong	1,252 tier 1
B06	What materials are used in this model?	elements by Material / Structural Material: • <By Category>: 350 • Softwood, Lumber: 198 • Aluminum: 96 • Metal Stud Layer: 34 • Metal - Steel - 345 MPa: 31 • Metal - Steel - 250 MPa: 28 • Concrete - Cast-in-Place Concrete: 26 • S…	Materials with counts. Lists Revit's placeholder "<By Category>" (350) as if it were a material.	Partial	153 tier 0
B07	Show me all elements belonging to the Walls category.	Found 9 Walls elements.	9 walls.	Correct	1,065 tier 1

Property-based queries

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
P01	Find all walls with a thickness greater than 200 mm.	Found 0 Walls elements where Thickness > 200.	9 walls: all are 280 mm thick (stored as Construction.Width). Answered 0: compared a "Thickness" property walls don't have, instead of their Width.	Wrong	156 tier 0
P02	Show all structural columns made of concrete.	Found 1 Structural Columns elements where Structural Material is "Concrete, Cast-in-Place gray".	1 concrete column (Concrete, Cast-in-Place gray).	Correct	153 tier 0
P03	Find all elements with a Mark value of C1.	The model has Mark values such as Pc-2.1, Pl-2.1, and S-3.3, but no C1. Did you mean one of the listed Mark values?	No element has Mark C1. Said so and offered real Mark values.	Correct	5,247 tier 2
P04	Show all beams with a length greater than 5 metres.	Found 49 Structural Framing elements where Length > 5000.	49 framing members longer than 5 m.	Correct	156 tier 0
P05	Find all elements with missing or empty Mark values.	The model lists Mark values for Structural Foundations, Walls, and Floors, but does not indicate which values are missing or empty. Would you like to find elements with a specific Mark value, or check one of these categories for b…	Missing Mark: 0/30 columns, 0/370 framing, 3/5 floors, 20/36 foundations, 4/9 walls have none. Claimed the data can't show missing values; it can.	Asked back	5,305 tier 2

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
P06	List all elements with a fire rating of 2 hours or more.	This model has no fire-rating property. Would you like to list elements by another available property, such as material, level, or thickness?	The model has no fire-rating property. Said so honestly.	Correct	5,089 tier 2
P07	Find all elements whose comments contain the word "review".	I couldn't find "whose" in this model, so I can't answer that exactly. Ask without it, or use a name from the model.	0: only one Comments value exists ("-"). Refused because of the word "whose" ("couldn't find 'whose'").	Wrong	5,157 tier 2
P08	Show all structural elements located on Level 2.	Found 0 elements where Family is "Structural Area Reinforcement" and Level is "Level 2".	92: 84 framing + 8 columns on Level 2. "structural" matched the family "Structural Area Reinforcement" -> 0 found.	Wrong	155 tier 0

Terminology

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
N01	Find all concrete columns.	Found 1 Structural Columns elements where Structural Material is "Concrete, Cast-in-Place gray".	1 concrete column.	Correct	150 tier 0
N02	Show me all vertical structural members.	Found 30 elements where Column Style is "Vertical".	30 columns (Column Style = Vertical). Good semantic mapping.	Correct	1,117 tier 1
N03	Find all horizontal structural members.	No elements where Family is "Structural Area Reinforcement" found.	370 structural framing members (beams, joists, braces). "structural" matched the Area Reinforcement family -> nothing found.	Wrong	151 tier 0
N04	Show me all floor slabs.	Found 5 Floors elements.	5 floors.	Correct	151 tier 0
N05	Find all foundations, including isolated footings and foundation walls.	I couldn't find "isolated" in this model, so I can't answer that exactly. Ask without it, or use a name from the model.	36 structural foundations (+ 9 walls based on the Foundation level). Refused because of the word "isolated".	Wrong	5,122 tier 2
N06	Show me all steel members.	Found 71 elements where Structural Material is one of "Metal - Steel - 250 MPa", "Metal - Steel - 345 MPa", "Steel, 45-345".	71 steel elements across categories.	Correct	151 tier 0
N07	Find all elements that could be considered part of the building's structural system.	Found 845 elements. Showing the first 500.	Structural categories (framing, columns, foundations, rebar, area reinforcement, floors, walls); exclude Lines, Runs, Supports. Returned all 845 elements without explaining the classification.	Partial	1,202 tier 1

Multi-condition

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
C01	Find all concrete columns on Level 2 with a cross-section larger than 400 × 400 mm.	The model has no cross-section size property for columns. Would you like to filter by a different available property, such as Structural Material or Top Level?	0: the only concrete column (450x450) is on Level 1, not Level 2. Said there is no cross-section property; a 450 mm "b" dimension exists.	Partial	5,617 tier 2
C02	Show all beams made of steel or concrete with a length greater than 6 metres.	The listed Structural Framing materials include steel but no concrete. Would you like to show steel beams only, or use a different category for concrete elements?	No concrete framing exists; steel framing longer than 6 m should be listed. Correctly noted no concrete framing, but asked back instead of answering the steel part.	Partial	5,240 tier 2

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
C03	Find all walls on Levels 1 and 2 that are thicker than 200 mm.	I couldn't find Walls with Level "Level 1" in this model. Options: Foundation.	None: all 9 walls are based on the Foundation level. Honest: listed the only level walls use.	Correct	5,359 tier 2
C04	Find all structural columns that are either concrete or steel, but exclude columns with missing Mark values.	I can filter columns by Structural Material, but this model does not list Mark for Structural Columns. Should I find concrete or steel columns without excluding columns with missing Mark values?	29 concrete/steel columns; columns carry no Mark (only Column Location Mark). Asked back; correctly noted columns have no Mark.	Partial	5,481 tier 2
C05	Find all elements with a volume greater than 1 m³ and a material containing concrete.	Found 14 elements where Volume > 1 and Structural Material contains "concrete".	14 concrete elements with volume > 1 m3.	Correct	1,255 tier 1
C06	Show all structural framing elements on Level 2 except those belonging to type W12x26.	I can show Structural Framing on Level 2, but this model has no W12x26 type. Did you mean a different type, or should I show all framing on Level 2?	Type W12x26 doesn't exist in the model. Said so and offered the alternative.	Correct	5,462 tier 2
C07	Find all elements with a height between 3 and 5 metres.	Found 4 elements where Unconnected Height >= 3000 and Unconnected Height <= 5000.	4 walls (Unconnected Height 4,700 mm).	Correct	1,148 tier 1

Statistics & quantities

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
A01	How many concrete columns are in the model, grouped by level?	Structural Columns elements where Structural Material is "Concrete, Cast-in-Place gray" by Level: • Level 1: 1	Level 1: 1 concrete column.	Correct	158 tier 0
A02	Calculate the total volume of concrete in all structural columns.	Total Volume of the 1 Structural Columns elements where Structural Material is "Concrete, Cast-in-Place gray": 0.607 m³.	0.607 m3.	Correct	156 tier 0
A03	Calculate the total length of structural framing elements by type.	Total Length of the 370 Structural Framing elements: 1,275,952.5 mm (1,275.953 m). • SC15012: 312,000 mm (312 m) • Timber_50 x200: 237,000 mm (237 m) • Timber_50 x150 _ContentGenerator: 194,659.4 mm (194.659 m) • 200UB25.4: 183,44…	Total framing length by type (SC15012 312 m, Timber_50 x200 237 m, …).	Correct	1,184 tier 1
A04	What are the five most common wall types in this model?	Which property should I rank by?	Only one wall type: CL_W1 (9 walls). Asked which property to rank by.	Asked back	1,091 tier 1
A05	Calculate the total volume of concrete for each level.	Total Volume of the 17 elements where Structural Material contains "Concrete": 215.875 m³. 32 more have no Volume and are not included. • Level 1: 111.217 m³ • Foundation: 90.094 m³ • (no Level): 14.564 m³	Concrete 215.875 m3 in total, split by level.	Correct	5,658 tier 2
A06	Which material has the highest total volume in the model?	Total Volume of the 418 elements: 234.942 m³. 427 more have no Volume and are not included. • Concrete - Rough: 110.61 m³ • CL Concrete_ panels: 90.094 m³ • Concrete - Cast-in-Place Concrete - 35 MPa: 14.564 m³ • <By Category>: 12…	Concrete - Rough, 110.61 m3. Shown as the first line of a breakdown, not stated as "the highest".	Correct	0 tier 0
A07	What is the average length of structural beams?	Average Length of the 370 Structural Framing elements: 3,448.52 mm (3.449 m).	3.449 m average. Correct, but used tier 2 (5,176 tokens) because of the word "structural".	Correct	5,176 tier 2
A08	Give me a breakdown of structural elements by category and material.	elements where Family is "Structural Area Reinforcement" by category: • Structural Area Reinforcement: 44	Category x material breakdown of the structural elements. "structural" bug: only the 44 area reinforcements.	Wrong	157 tier 0

Spatial

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
S01	Show me all structural columns on Level 1.	Found 1 Structural Columns elements where Level is "Level 1".	1 column (base Level 1).	Correct	155 tier 0
S02	Find all elements located within 2 metres of Grid A.	This model has no grid location or distance-to-grid property. Do you mean elements that move with grids, or elements on a particular level?	Not possible: grids are not in the data. Said so.	Honest "can't"	5,167 tier 2
S03	Identify the elements located at the highest elevation in the model.	6 elements share the highest Elevation at Top of the elements: 9,044.45 mm (9.044 m). (434 of them have no Elevation at Top and were not ranked.)	6 framing members at the highest top elevation, 9.044 m.	Correct	1,188 tier 1
S04	Find all columns that extend between Level 1 and Level 2.	Found 1 Structural Columns elements where Level is "Level 1" and Top Level is "Level 2".	1 column from Level 1 to Level 2.	Correct	1,189 tier 1
S05	Show all structural elements within a 5-metre radius of this selected column.	Which column do you mean? I need its element ID or location. This model also has no element coordinates or distance property to determine which structural elements are within 5 metres.	Not possible: the app can't see the viewer selection. Asked for an element id; wrongly said there are no coordinates (positions exist).	Honest "can't"	5,287 tier 2
S06	Identify elements that intersect or overlap other structural elements.	This model has no property for detecting intersections or overlaps. Would you like to identify elements by another available property, such as level, material, or family/type?	Not possible: no clash detection. Said so.	Honest "can't"	5,408 tier 2
S07	Show all elements within the northern half of the building.	This model has no location or building-half property to identify the northern half. Can you specify a level, category, or another available property to use?	Not possible: no north/orientation data. Said so.	Honest "can't"	5,152 tier 2
S08	Which structural elements are closest to the selected beam?	I couldn't find Structural Framing with Family "Structural Area Reinforcement" in this model. Options: Rectangular(Timber)-Beam, Light Gauge-Joists, UB-Universal Beams (AS 3679_1), M_TJL Wood Open Web Joist, CHS-Circular Hollow Se…	Not possible: the app can't see the selected beam. Confusing message about the Area Reinforcement family.	Wrong	0 tier 0

Relationships

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
R01	Which beams are connected to column C1?	I can find beams, but this model has no column mark C1. Could you provide a listed column mark, such as C-6, C-7, or C-8?	Connections are not in the data; C1 doesn't exist. Said C1 doesn't exist, offered real marks; didn't mention connectivity is unavailable.	Honest "can't"	5,184 tier 2
R02	Which columns support the beams on Level 2?	Found 8 Structural Columns elements where Level is "Level 2".	Support relationships are not in the data. Listed the 8 columns on Level 2 as if they were the supports: misleading.	Wrong	1,222 tier 1
R03	Find all walls connected to the selected floor slab.	Which floor slab do you mean? Please provide its element ID or another identifying detail; the model has no selected-element information.	Not possible: no connectivity, no viewer selection. Asked which slab.	Honest "can't"	5,189 tier 2
R04	Which elements belong to the same Revit family type as this selected element?	Which selected element do you mean? Please provide its name or element ID.	Not possible without knowing the selected element. Asked for the element.	Honest "can't"	1,528 tier 1
R05	List all elements belonging to the same assembly as this element.	Which element do you mean? Please provide its name or ID.	Assemblies are not in the data / no selection. Asked for the element.	Honest "can't"	1,228 tier 1
R06	Find all structural framing elements supported by columns on Level 2.	Found 84 Structural Framing elements where Level is "Level 2".	Support relationships are not in the data. Listed all 84 framing members on Level 2 as if "supported by columns": misleading.	Wrong	1,217 tier 1

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
R07	Which elements share the same material and type as this selected element?	Which element do you mean? Please provide its element ID so I can find elements with the same material and type.	Not possible without knowing the selected element. Asked for the element id.	Honest "can't"	6,157 tier 2

Conversation

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
I01	Show me all structural columns on Level 1.	Found 1 Structural Columns elements where Level is "Level 1".	1 column on Level 1.	Correct	155 tier 0
I02	Now show me only the concrete ones.	Found 1 Structural Columns elements where Level is "Level 1" and Structural Material is "Concrete, Cast-in-Place gray". follow-up	1 (it is the concrete column). Follow-up kept the context, 0 tokens.	Correct	0 tier 0
I03	Exclude columns with a Mark value of C1.	The model has no Mark property for Structural Columns. Did you mean a different column property, such as Column Location Mark?	Columns have no Mark value. Said so and suggested Column Location Mark.	Honest "can't"	5,275 tier 2
I04	How many elements remain?	What should the elements remain after—are you asking for the total number of elements in the model, or a count after applying a filter?	1 element remains. Lost the conversation context and asked back.	Wrong	5,074 tier 2
I05	Highlight these elements in the 3D viewer.	Found 1 Structural Columns elements where Level is "Level 1" and Structural Material is "Concrete, Cast-in-Place gray". follow-up	The same 1 column, highlighted.	Correct	1,213 tier 1
I06	Add these results to the Schedule of Answers.	I can show the matching concrete structural columns, but I can't add results to a Schedule of Answers or highlight elements in the 3D viewer. Would you like me to list the matching columns? follow-up	Every answer is already added to the Schedule of Answers automatically. Said it can't add to the schedule or highlight, which is not true.	Partial	5,598 tier 2
I07	Compare this selection with all structural columns on Level 2.	What would you like to compare the selection by?	Comparison of selections is not supported. Asked what to compare by.	Asked back	1,209 tier 1
I08	Clear the current selection and find all concrete walls.	Found 9 Walls elements where Structural Material contains "Concrete".	9 concrete walls, starting afresh.	Correct	1,195 tier 1

Stress & ambiguity

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
E01	Find all concrete columns with a length of 100 metres.	Found 0 Structural Columns elements where Structural Material contains "Concrete" and Length is 100000.	0: no invented elements.	Correct	1,159 tier 1
E02	Find all elements made of unobtainium.	Unobtainium isn't listed as a material in this model. Which available material should I search for?	Unobtainium doesn't exist. Said so and offered real materials.	Correct	5,064 tier 2
E03	Show me all columns on Level 99.	There is no Level 99 in this model. Which available level should I use: Foundation, Level 1, Level 2, Level Lower, or Ref Beam system?	Level 99 doesn't exist. Listed real levels (missed Top of Parapet and Ceiling).	Correct	5,328 tier 2
E04	Find columns with a width greater than 300 mm.	Found 0 Structural Columns elements where Width > 300.	1: the concrete column is 450 mm wide; steel widths are 133 mm and 63.5 mm. Answered 0: steel widths are stored in cm (13.3 cm) and were compared with 300 directly; the 450 mm "b" was missed.	Wrong	155 tier 0
E05	Find all large beams.	What should "large" mean—longer than a specified length, or a particular beam size/type?	Ask what "large" means. Asked for a threshold, as expected.	Correct	5,079 tier 2

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
E06	Find all columns with a cross-section of 400 × 400 mm, regardless of how their dimensions are stored.	I can't find a 400 × 400 mm cross-section property for columns in this model. Would you like to search by a listed column type or section name instead?	0: no 400x400 column (the concrete one is 450x450). No invented elements, but said no cross-section data exists.	Partial	5,369 tier 2
E07	Find all concrete columns, but ignore elements with incomplete material information.	I can find concrete columns using Structural Material. What do you mean by "incomplete material information"?	1 concrete column; no column has incomplete material information. Asked what "incomplete" means instead of answering.	Partial	5,323 tier 2
E08	Find all elements that are not structural.	Found 801 elements where Family is not "Structural Area Reinforcement". Showing the first 500.	Non-structural: Lines, Runs, Supports, Stairs (5 elements). "structural" bug: returned 801 elements that aren't area reinforcement.	Wrong	1,087 tier 1

Engineering workflows

ID	Question	Augmented's answer	Right answer (checked in the model data)	Result	Tokens
W01	Prepare a quantity takeoff of all concrete columns, grouped by level and type.	Structural Columns elements where Structural Material is "Concrete, Cast-in-Place gray" by Level / Type: • Level 1: 1 • 450 x 450mm: 1	1 concrete column, Level 1, type 450 x 450mm, 0.607 m3. Counts by level and type, but no volumes.	Partial	1,371 tier 1
W02	Identify all structural elements with missing material assignments.	Which structural categories should I check? The model has Structural Framing, Structural Rebar, Structural Area Reinforcement, Structural Foundations, and Structural Columns. Material properties are available for some categories, …	390: all 346 rebar and 44 area reinforcement have no structural material. Asked which categories to check.	Asked back	5,602 tier 2
W03	Find elements with missing or duplicate Mark values.	I can check Mark values for Structural Foundations, Walls, and Floors, but the model data doesn't indicate which Marks are missing or duplicated. Would you like me to list the elements with Mark values so you can review them?	Missing Marks per category (see P05); duplicate check not supported. Claimed the data can't show missing values.	Asked back	5,266 tier 2
W04	Generate a structural element inventory showing category, type, level, material and Mark.	Found 845 elements. Showing the first 500.	A table with category, type, level, material and Mark. Listed the elements, but the table has no material/Mark columns (no export).	Partial	1,705 tier 1
W05	Identify potential inconsistencies between elements of the same type.	What kind of inconsistency should I check for between elements of the same type—for example, differences in a particular property?	Not supported yet. Asked what kind of inconsistency.	Asked back	1,382 tier 1
W06	Prepare a list of all structural framing elements with their lengths and materials.	Found 370 Structural Framing elements.	370 framing members with length and material. Listed them, but the table shows no length/material columns.	Partial	5,333 tier 2
W07	Find all elements with no assigned level or an invalid level reference.	Do you mean elements with a blank level value, or elements whose level reference does not match an existing level?	Elements without a level property (e.g. rebar, lines). Asked back.	Asked back	1,215 tier 1
W08	Produce a summary of the structural model suitable for a preliminary engineering review.	What would you like included in the structural model summary—for example, counts by category, materials, levels, or member sizes?	A grounded model summary. Asked what to include.	Asked back	5,266 tier 2